

A Project report
on
VacciCare+

Submitted To:

**G H Raison International Skill Tech University
PUNE– 411006**

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Submitted By:

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**Semester III
Year2025-26**

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PUNE-411006

CERTIFICATE

This is to certify that Ms. Kanishka Patil and Ms. Prerna Fartade, student of SYBCA, Roll No:BC1048 and BC1013 respectively, has satisfactorily completed their Project “**VacciCare+**” and submitted the report for the academic Year 2025-26.

Date: _____

Project Guide
Prof.Vaishnavi Lokhande

Internal Examiner

External Examiner

H O D

Dean

ACKNOWLEDGEMENT

A Successful Project is the result of a good team-work Which consists of the not only the Partners Who Put their long and hard work but also those who guided them.

Indeed, a true saying. First of all , I would like to thanks, our Honourable **Vice Chancellor** to give me an opportunity to study in this institute.

I am also grateful to **Ms. Vaishnavi Lokhande** for providing necessary information, valuable cooperation, encouragement and all the technical help during the completion of my project I am also thankful to all faculty members who has supporting me to complete my Project.

Last but not least, we express our gratitude to the almighty without whose blessing nothing is possible.

Signature of Student's:

Kanishka Patil

Prerna Fartade

ABSTRACT

VacciCare+ is a digital healthcare management system designed to simplify and enhance the process of vaccination tracking, scheduling, and record management. The platform provides a secure and user-friendly interface for healthcare administrators, doctors, and patients to manage vaccination details efficient.

The system enables healthcare providers to register patients, maintain digital vaccination records, send automated reminders for upcoming doses, and generate vaccination certificates. Patients can easily view their vaccination history, receive timely notifications, and stay updated on their immunization schedule.

Developed using [mention your technologies, e.g., Angular/React for frontend, Java/Python for backend, MySQL for database], VacciCare+ ensures data security, scalability, and accessibility across devices. By digitizing manual vaccination processes, the system reduces errors, improves efficiency, and provides real-time access to critical health information.

In a world where timely immunization is crucial for public health, VacciCare+ offers a modern, reliable, and technology-driven solution to ensure better healthcare outcomes.

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INTRODUCTION:

Vaccination is one of the most effective and affordable methods of preventing life-threatening diseases in children. Despite its importance, many healthcare centers in developing countries still rely on manual, paper-based systems to manage vaccination records. These traditional methods often result in challenges such as missing or misplaced records, delayed immunizations, human errors in calculating due dates, and difficulties in generating reports or sending timely reminders to parents.

To overcome these issues, VacciCare+ is designed as a web-based Child Vaccination Management System. Developed using Python (Flask/Django) for the backend and MySQL for secure data storage, the system integrates a simple yet responsive frontend built with HTML5, CSS (Bootstrap), and JavaScript. The platform allows parents to register their children, manage profiles, and book vaccination appointments with just a few clicks. Hospitals can update vaccination status, while administrators oversee records, approvals, and reports.

The system also incorporates smart features such as automated vaccination reminders, QR code-based verification, and search and filter capabilities to enhance usability. By digitalizing vaccination management, VacciCare+ not only improves record-keeping and reduces manual effort but also ensures timely immunizations—contributing to improved child health outcomes and stronger public health systems.

PROBLEM DEFINATION :

The E-Vaccination Management System is designed to digitalize ,In India, due to inadequate healthcare facilities and lack of awareness, many children miss their scheduled vaccinations. This system aims to ensure timely vaccination by maintaining accurate records, generating reminders, and providing accessible information to parents and healthcare workers.

The proposed system allows parents to register their children through a web or mobile platform. Based on the child's date of birth, the system automatically generates a personalized vaccination schedule and sends notifications for upcoming or overdue doses. Parents can also search for nearby hospitals and book vaccination appointments, ensuring convenience and timely care.

Existing System

In many healthcare centers, child vaccination records are maintained manually using paper-based registers or basic spreadsheets (like Excel). Healthcare workers must manually calculate due dates based on the child's birth date, which is time-consuming and error-prone. There is no automated system to send reminders to parents or generate reports. Data retrieval for audits or follow-ups is difficult, especially when records are misplaced or damaged. Tracking vaccine inventory, batch numbers, and expiry dates is also done manually or not at all.

Need For New System

In many healthcare centers, child vaccination records are still maintained manually through paper-based registers or basic spreadsheets. This traditional approach presents several challenges:

1. Human Errors: Calculating vaccination due dates manually often leads to mistakes, increasing the risk of missed or delayed immunizations.
2. Data Loss: Paper records are vulnerable to damage, misplacement, or loss, making long-term tracking unreliable.
3. Lack of Reminders: Parents are not notified about upcoming or overdue vaccinations, which results in lower vaccination coverage.
4. Inefficient Reporting: Generating reports for audits, follow-ups, or healthcare planning is time-consuming and inaccurate.
5. Limited Accessibility: Parents and healthcare workers cannot easily access or share vaccination histories when needed.

Scope Of System:

The proposed VacciCare+ Child Vaccination Management System is designed to improve vaccination tracking and healthcare efficiency by digitalizing records and automating reminders. Its scope includes the following key areas:

1. Maintain digital records of children and their vaccination history.
 2. Automatically calculate due vaccinations based on date of birth.
 3. Allow health workers to record administered vaccines.
 4. Generate reports for due, missed, and completed vaccinations.
 5. Simple web-based system using Python (Flask) and SQLite.
 6. Optional features: inventory tracking, SMS/email reminders.
- Digital Record Maintenance : Securely store children's personal and vaccination data. Maintain complete vaccination history that can be accessed anytime.
 - Automated Vaccination Scheduling Generate vaccination schedules automatically based on the child's date of birth.
 - Reminder and Notification Services Send alerts for upcoming or overdue vaccinations through email, SMS, or system notifications
 - User Feedback : Allow users to submit feedback for continuous improvement of the system.

Feasibility Study

Before developing the VacciCare+ Child Vaccination Management System, a feasibility study was conducted to ensure that the proposed solution is practical, achievable, and beneficial. The study covers the following aspects:

1. Operational Feasibility : The system is designed to be user-friendly for parents, hospitals, and administrators. Parents can easily register, book appointments, and receive reminders. Hospitals can manage appointments and update vaccination status, while administrators oversee data and approvals. Since all stakeholders benefit directly, operational feasibility is strong.

2. Economic Feasibility : The system reduces dependence on paper-based records and manual tracking, lowering administrative costs for healthcare providers. Since the project relies on open-source tools, development costs are minimal. The long-term benefits of timely vaccination and reduced disease risk far outweigh the development and maintenance expenses.

3. Legal Feasibility : The system ensures secure handling of child and parent data, complying with privacy and ethical standards. Sensitive data is stored securely in the database, and access is restricted to authorized users only (parents, hospitals, admin).

4. Social Feasibility : Parents and healthcare workers face difficulties in managing vaccination records manually. A digital system that sends reminders, ensures timely vaccination, and simplifies record access is likely to be well-accepted by users. This enhances trust in healthcare and contributes to improved child health outcomes.

Hardware and Software Requirements

Hardware Requirements:

For User (Client Side):

- Processor: Intel i3 or higher / AMD equivalent
- RAM: 4 GB or more
- Storage: Minimum 250 GB HDD / 128 GB SSD
- Display: 1024 × 768 resolution or higher
- Network: Stable internet connection (Broadband / Wi-Fi)
- Device: Desktop, Laptop, or Smartphone (for accessing website)

For Server (Hosting Side):

- Processor: Intel i5 or higher, Quad-core (or equivalent cloud server CPU)
- Storage: 500 GB HDD / 256 GB SSD
- Network: High-speed internet with static IP or cloud hosting
- Backup Device: External hard drive or cloud backup storage

Software Requirements:

For User (Client Side) :

- Operating System: Windows 10 / 11, Linux, or macOS, Android/iOS for mobile
- Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest versions)

For Server Side / Development :

- Operating System: Windows Server / Linux (Ubuntu preferred for hosting)
- Front-End: HTML5, CSS3, JavaScript, Bootstrap
- Back-End: PHP / Python (Django/Flask) / Node.js (depending on choice)
- Database: MySQL / PostgreSQL
- Web Server: Apache / Nginx
- IDE / Editor: Visual Studio Code / PyCharm / Sublime Text
- Version Control: Git & GitHub/GitLab
- Other Tools: XAMPP/WAMP (if using local server), Postman (for API testing)

Requirement Analysis

Functional requirements : define the specific operations that the system must perform.

User Requirements :

- Register and create a new account.
- Login and logout securely.
- View available vaccines and schedules.
- Book vaccination appointments online.
- Access and download vaccination records.
- Receive notifications/reminders for upcoming vaccinations.
- Update personal profile information.

Admin Requirements :

- Secure admin login and authentication.
- Manage user accounts (approve, block, delete).
- Add, update, and delete vaccine details.
- Verify and update vaccination records.
- Generate reports on vaccination status and user activity.
- Monitor and maintain system data integrity.
- Store and retrieve vaccination records efficiently.
- Provide automated reminders via email/SMS.
- Allow multiple users to access the system simultaneously.

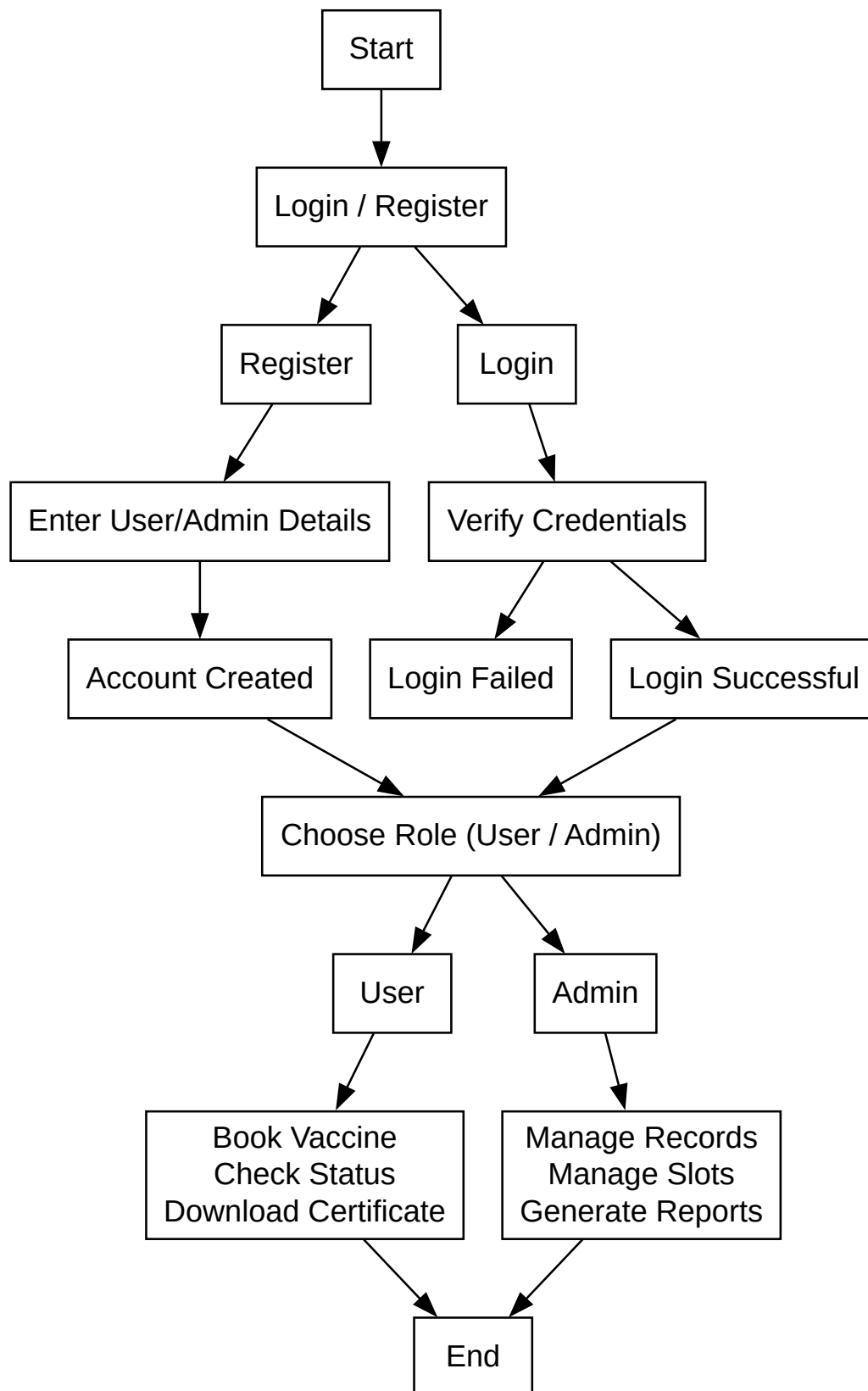
Non-Functional Requirements: Non-functional requirements describe the quality attributes of the system.

- **Performance:** The system should handle multiple concurrent users without performance issues.
- **Reliability:** Ensure 99.9% uptime for uninterrupted access.
- **Security:**
 - Encrypt sensitive data.
 - Provide role-based access control.
 - Use secure authentication methods.
- **Usability:**
 - Simple and user-friendly interface.
 - Responsive design for desktop, tablet, and mobile devices.
- **Scalability:** The system should support expansion (adding new vaccines,

Decision Table:

Rule #	Name	Who is Logged In	Hospital Approved?	Appointment Status	Allowed Actions
Rule 1	Parent Books	Parent	Yes	Pending	- Parent Books Appointment
Rule 2	Appointment View	Parent	Yes	Pending / Completed	- Parent Views Status
Rule 3	Hospital Approves/Updates	Hospital	Yes	Pending / Completed	- Hospital Views Appointment - Hospital Updates Status
Rule 4	Admin View/Export	Admin	Yes	Pending / Completed	- Admin Views Child Data & Appointment - Admin Exports Data
Rule 5	Hospital Not Approved	Parent / Hospital	Not Approved	-	- Hospital Requests Approval- Parent Cannot Book/View Appointment

Decision Tree:



Normalized Database Design (3NF):

User Table:

Field Name	Data Type	Description
user_id (PK)	INT (Auto	Unique ID for each
name	VARCHAR	Parent/guardian
email	VARCHAR	Email address
phone	VARCHAR	Contact number
address	VARCHAR	Address

Children Table:

Field Name	Data Type	Description
child_id (PK)	INT (Auto	Unique ID for each
user_id (FK)	INT	Reference to Users
name	VARCHAR	Child's name
dob	DATE	Date of birth
gender	VARCHAR	Male/Female/Other

Vaccination Table:

Field Name	Data Type	Description
vaccine_id (PK)	INT (Auto	Unique vaccine ID
name	VARCHAR	Vaccine name
description	TEXT	Detailed info about
dose_number	INT	Dose sequence
recommended_age	VARCHAR	Age or schedule

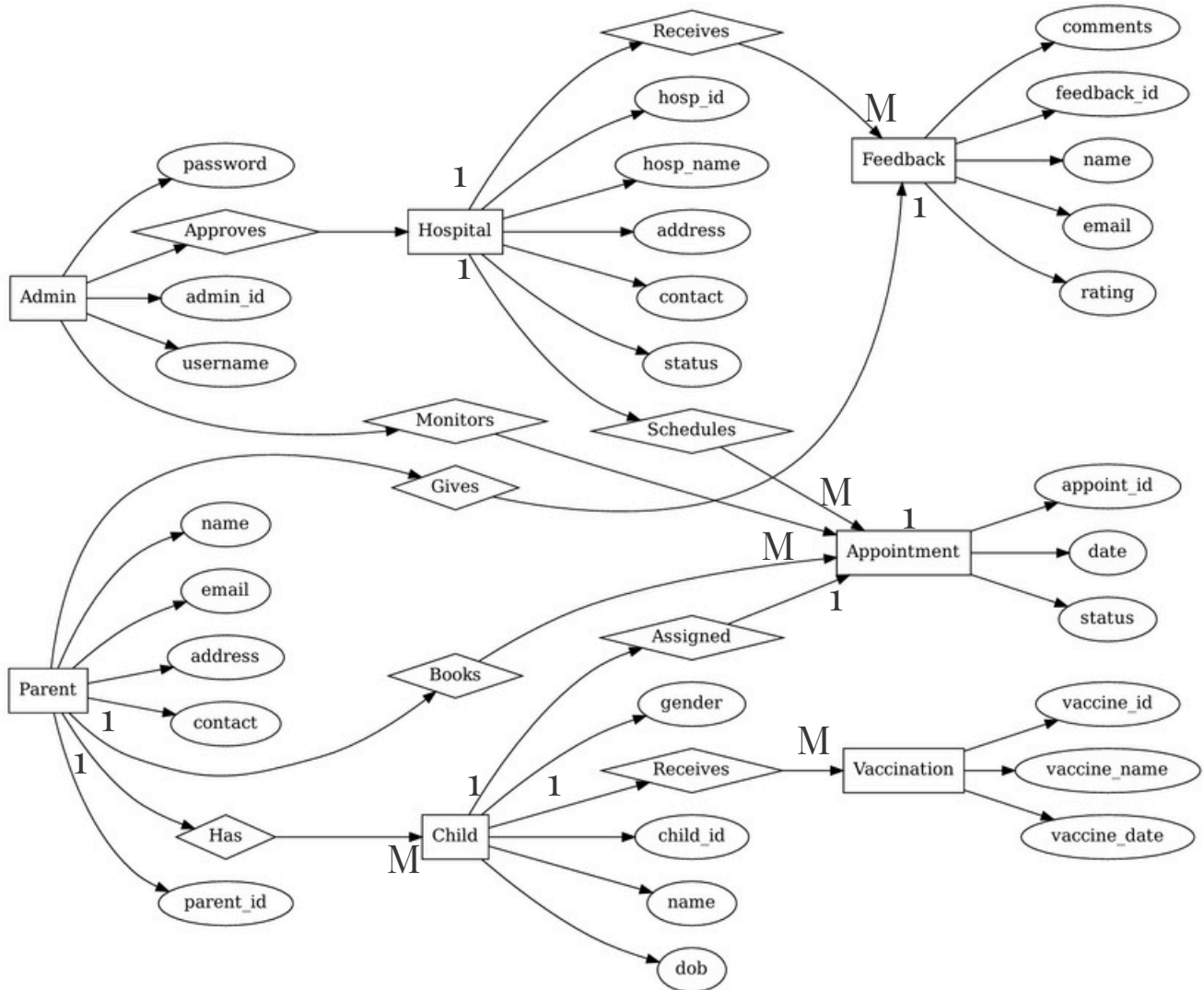
Vaccination Schedule Table:

Field Name	Data Type	Description
schedule_id (PK)	INT (Auto	Unique ID for
child_id (FK)	INT	Child receiving the
vaccine_id (FK)	INT	Vaccine to be given
scheduled_date	DATE	Scheduled date for
status	VARCHAR	Pending /

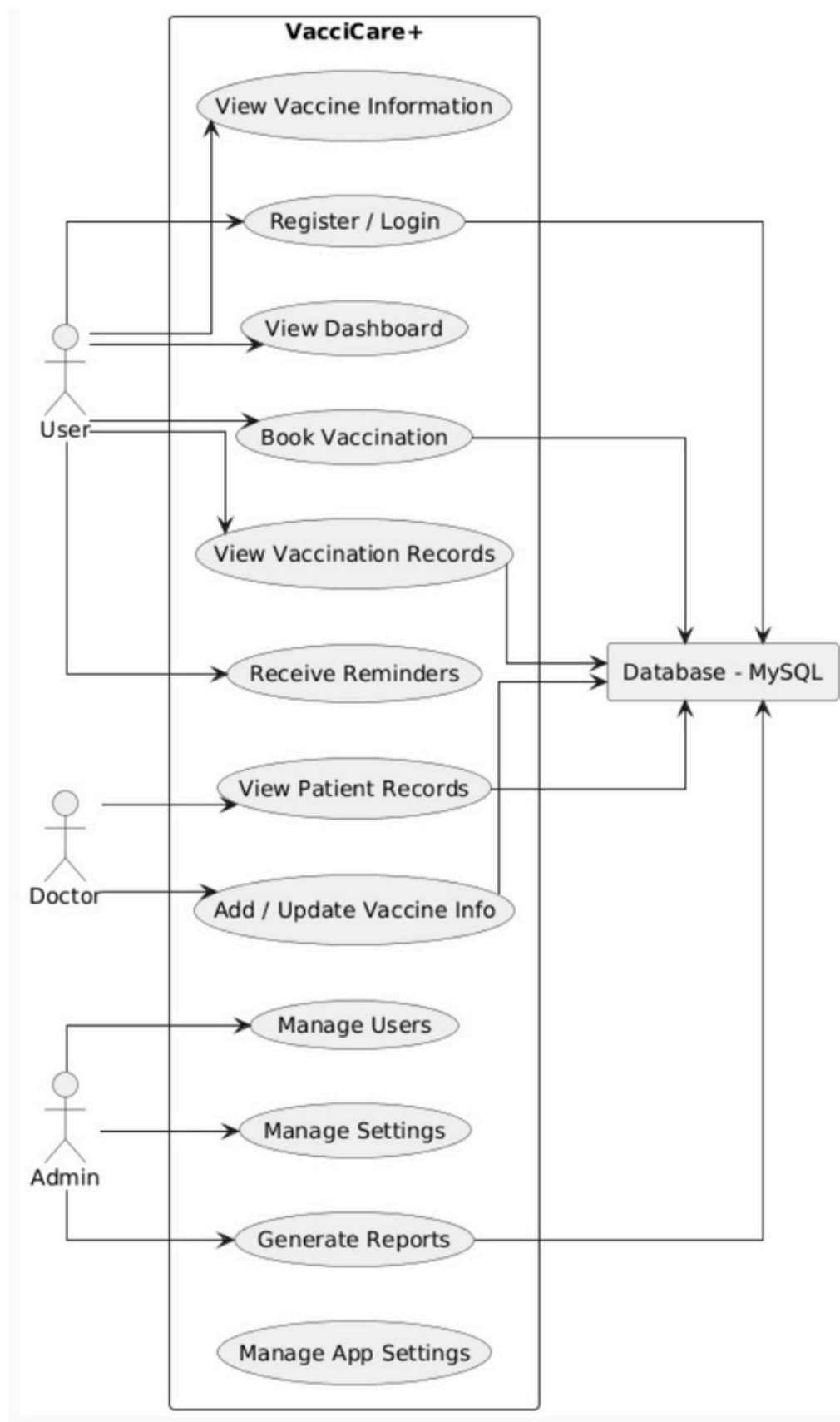
Admin Table:

Field Name	Data Type	Description
admin_id (PK)	INT	Unique admin ID
name	VARCHAR	Admin name
email	VARCHAR	Admin email
password	VARCHAR	Hashed password

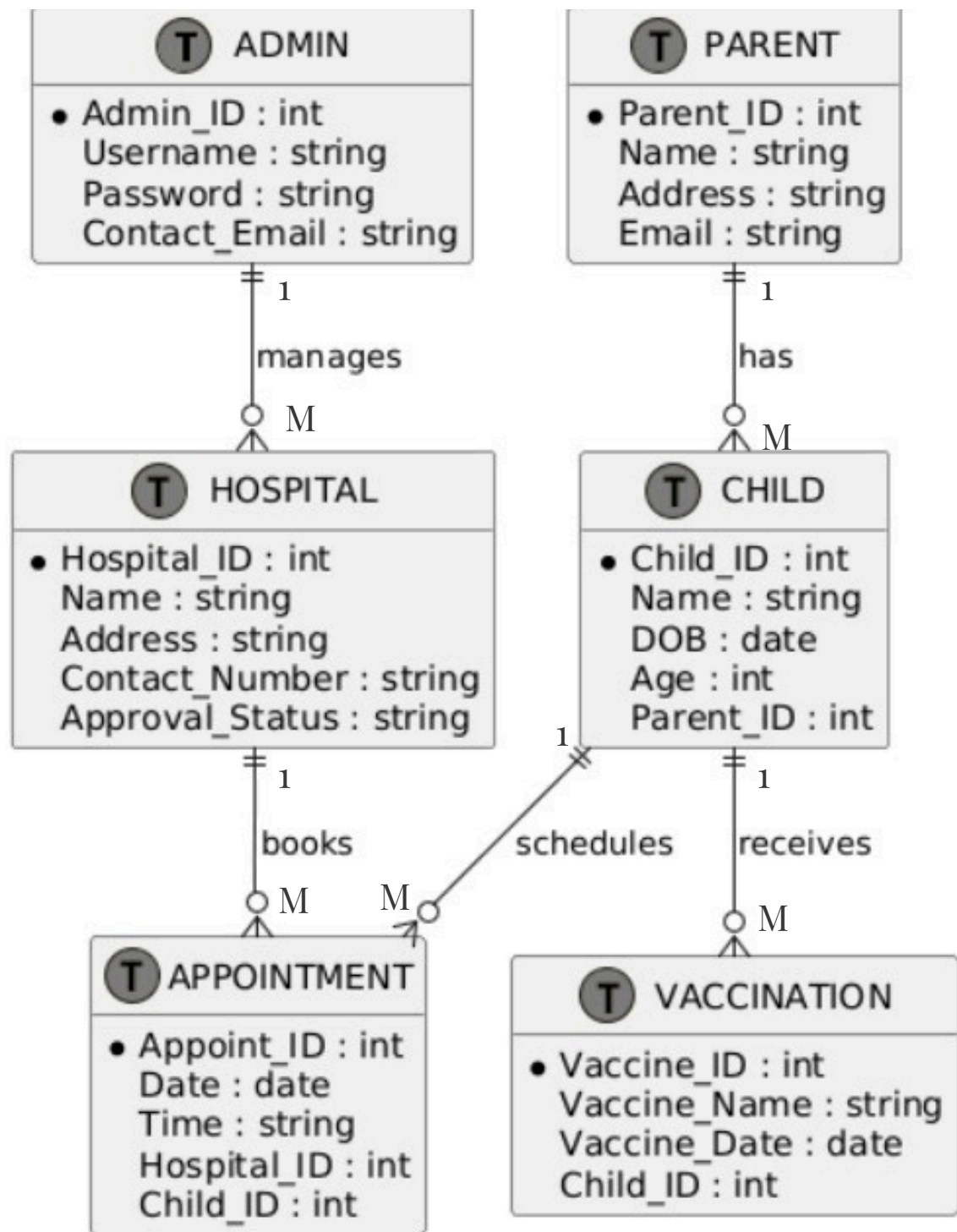
ER Diagram:



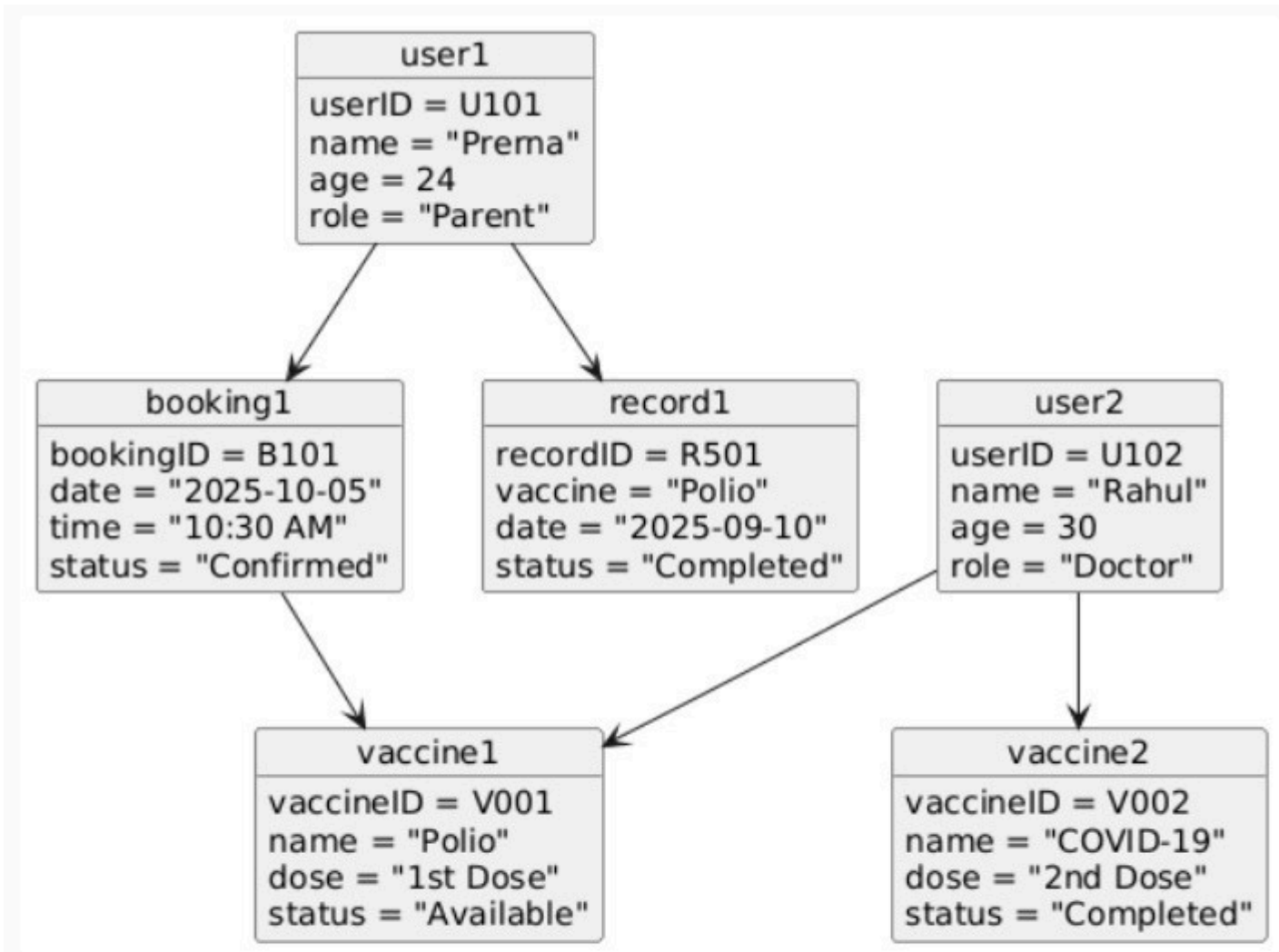
Use Case Diagram:



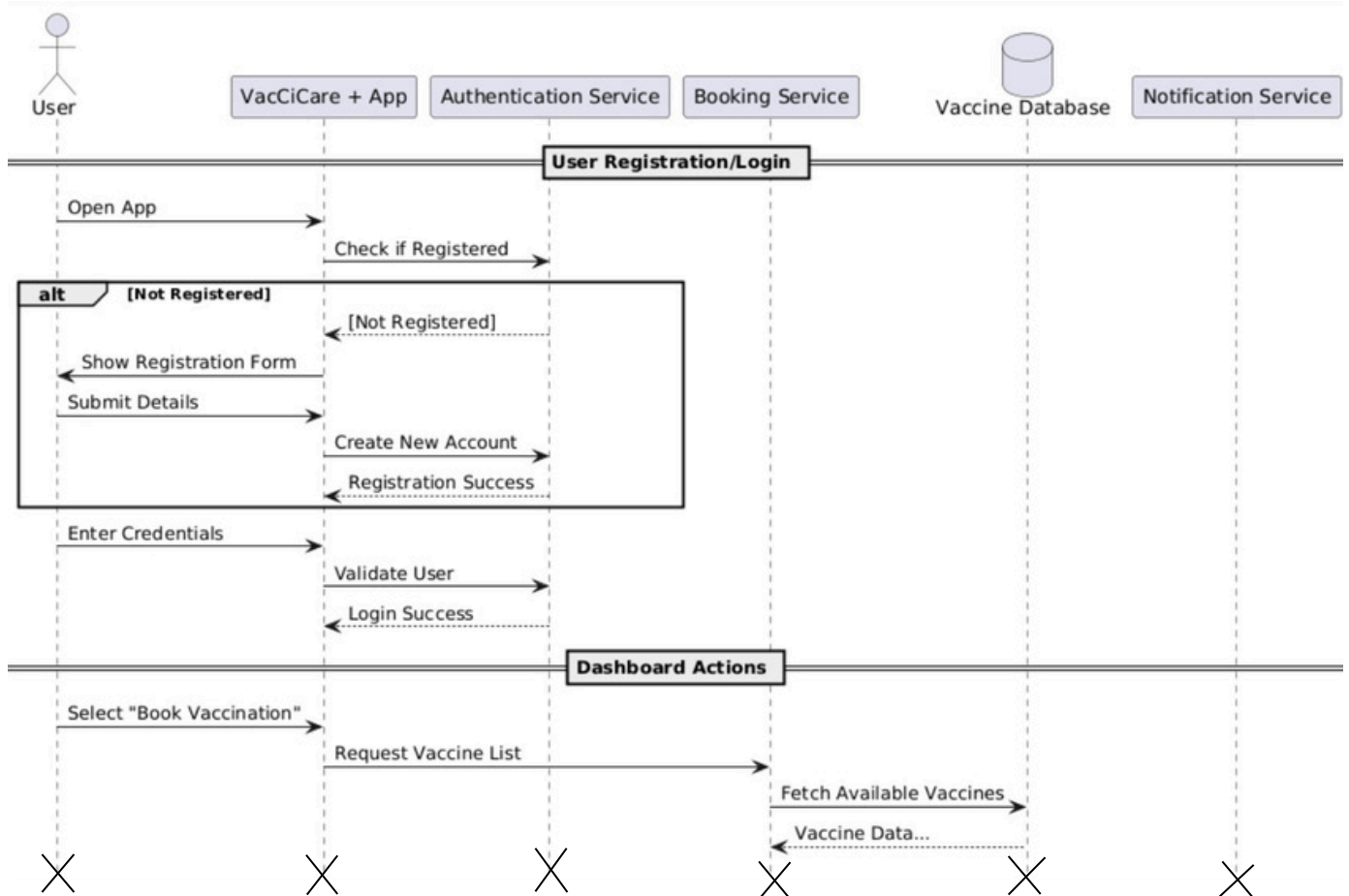
Class Diagram



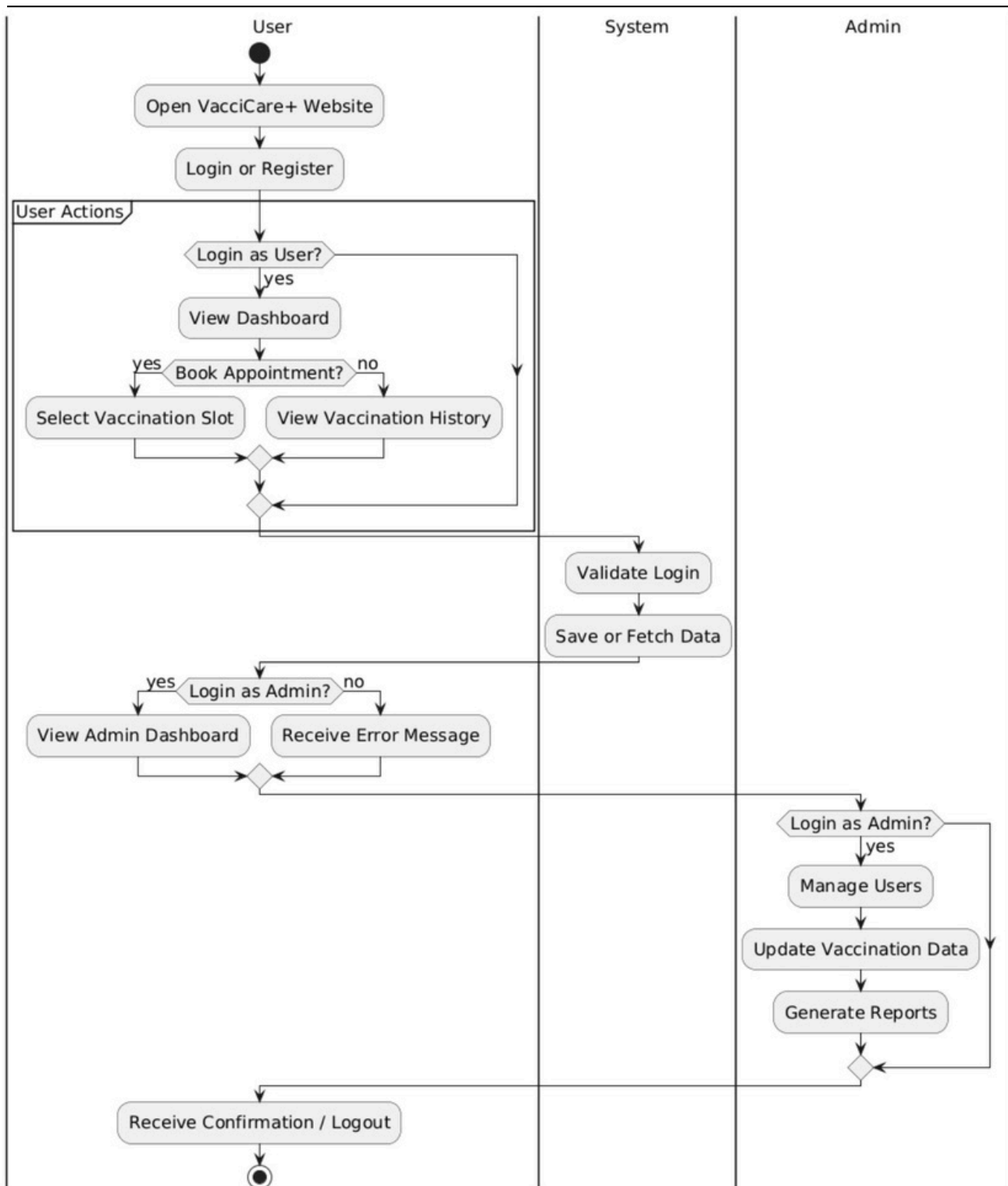
Object Diagram:



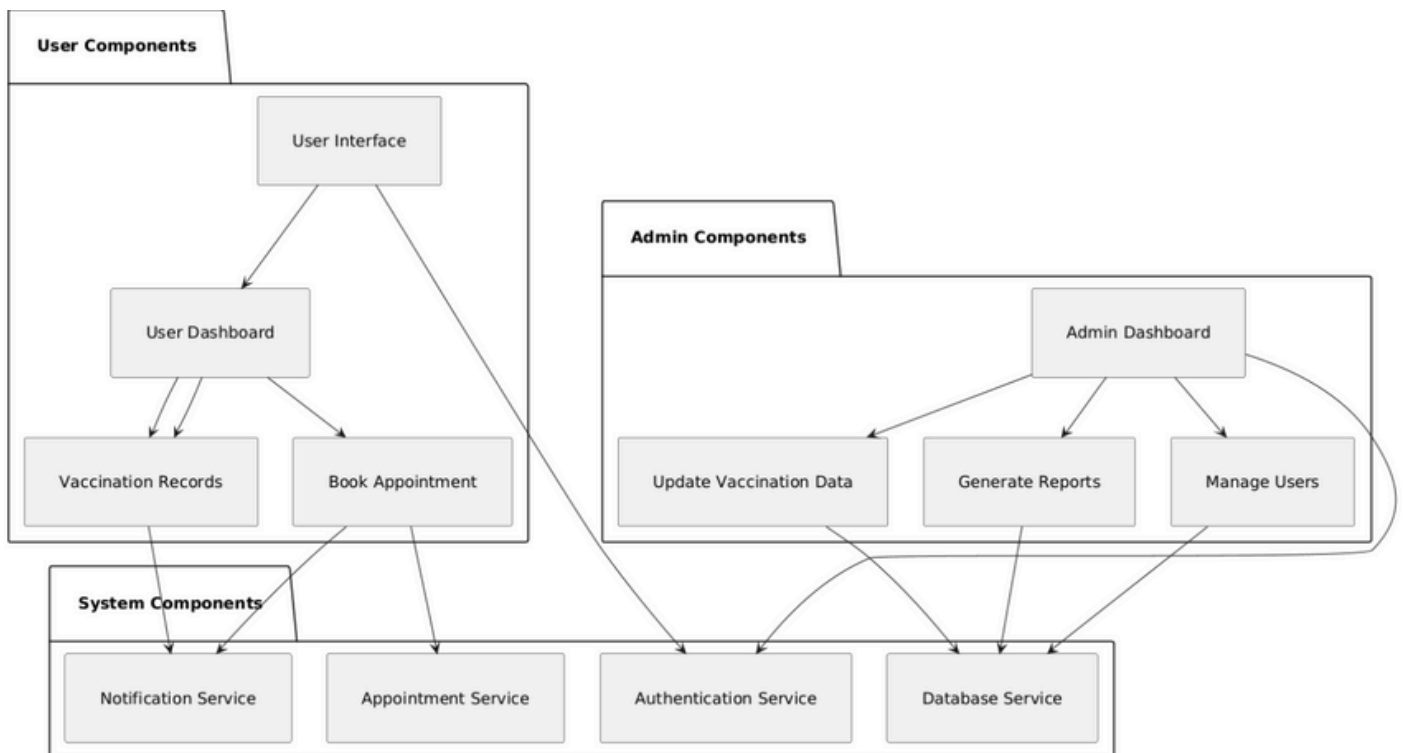
Sequence Diagram :



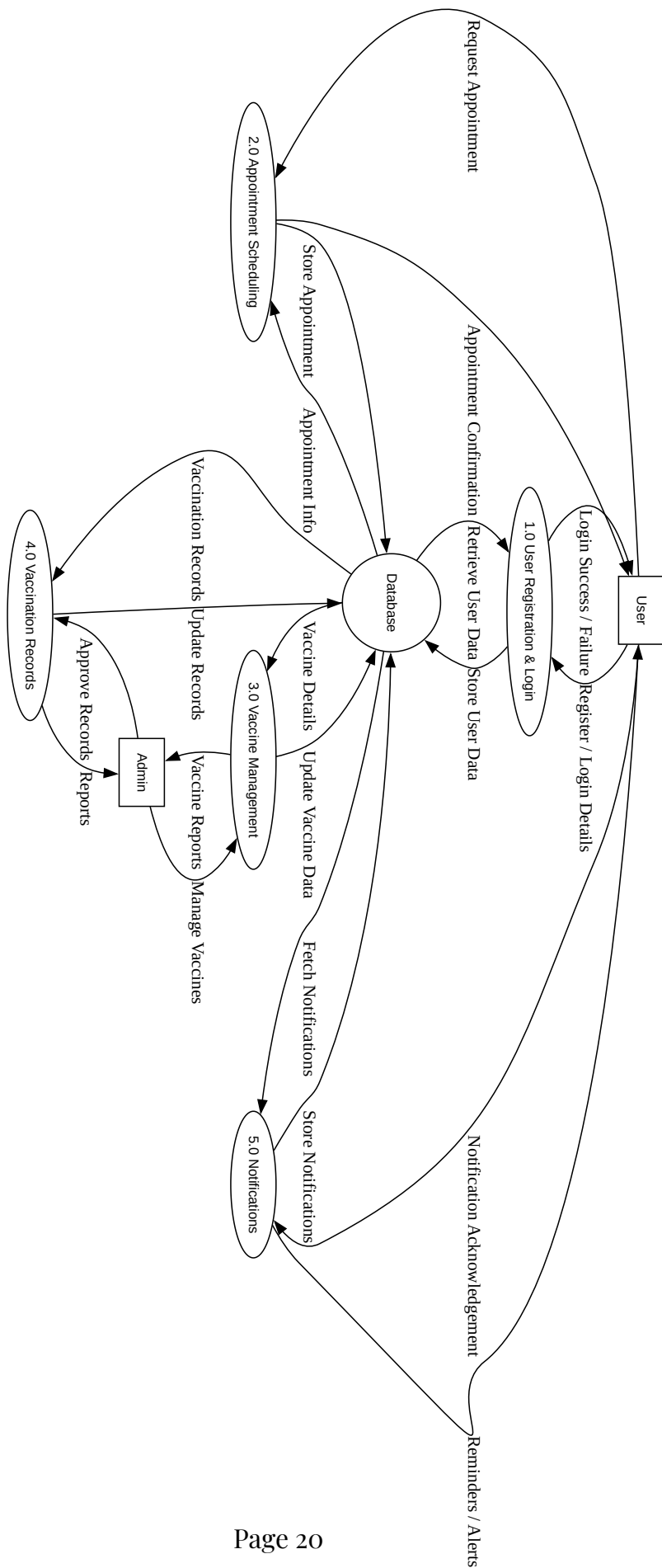
Activity Diagram:



Component Diagram :

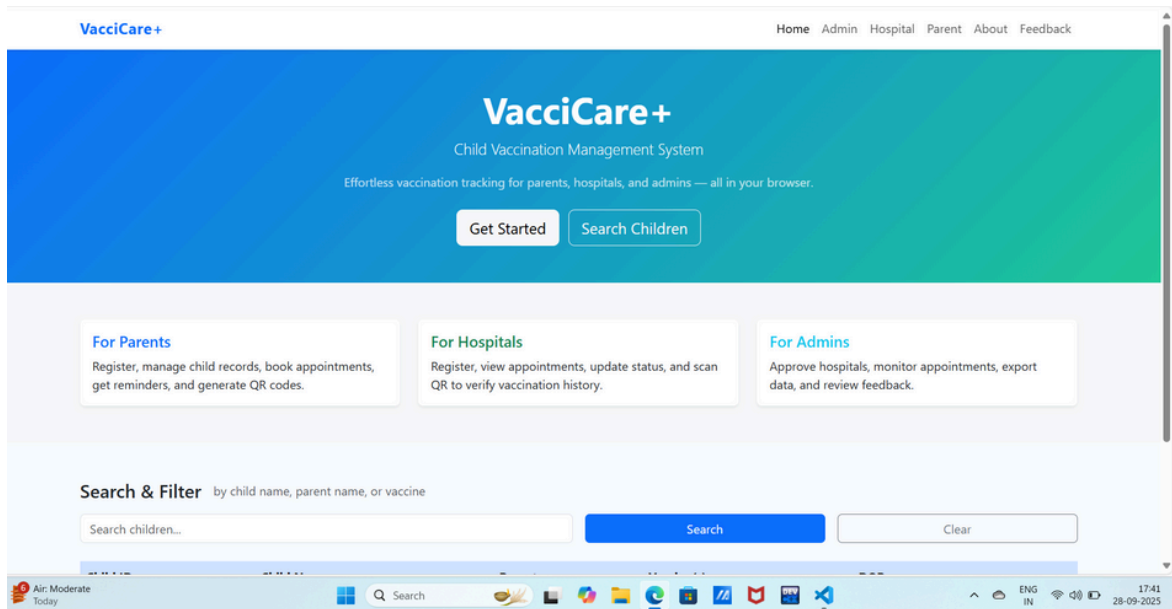


Data Flow Diagram :

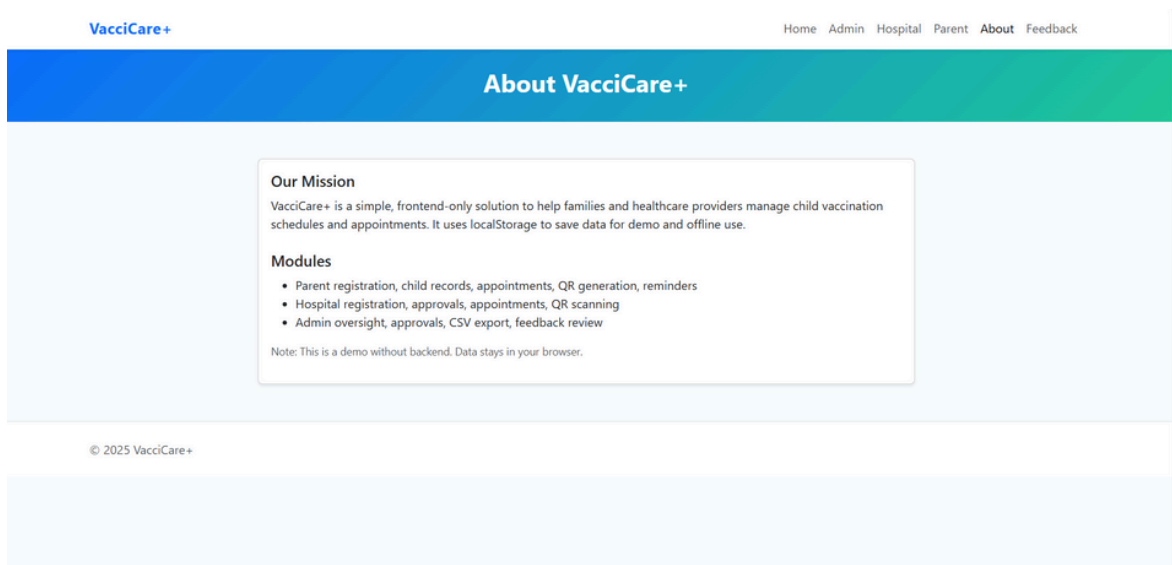


Input Screens:

Home screen

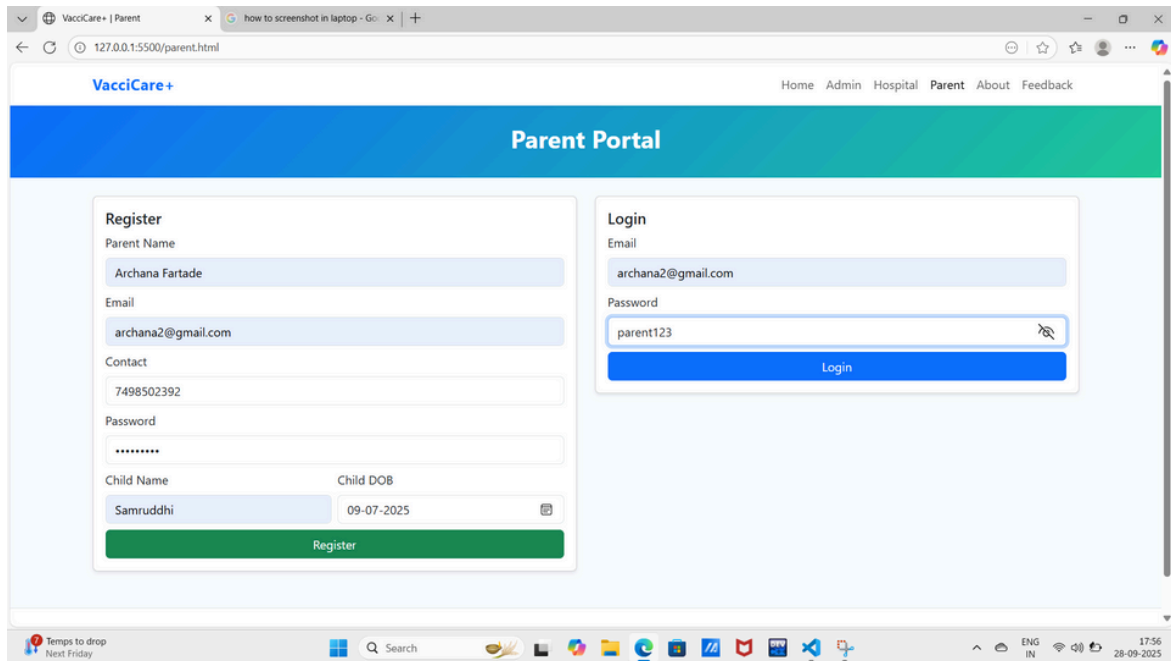


About VacciCare+:



User Module:

Registration Screen:



The screenshot shows a web browser window with the URL `127.0.0.1:5500/parent.html`. The page is titled "VacciCare+" and has a navigation bar with links: Home, Admin, Hospital, Parent, About, and Feedback. The main heading is "Parent Portal".

There are two main sections: "Register" and "Login".

Register Form:

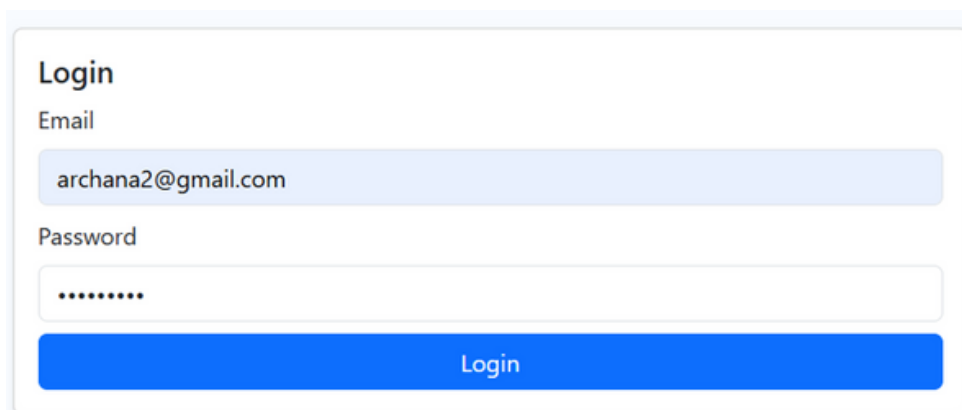
- Parent Name: Archana Fartade
- Email: archana2@gmail.com
- Contact: 7498502392
- Password: (masked with dots)
- Child Name: Samruddhi
- Child DOB: 09-07-2025
- Register button (green)

Login Form:

- Email: archana2@gmail.com
- Password: parent123
- Login button (blue)

The bottom of the browser window shows a Windows taskbar with various icons and a system tray indicating the time as 17:56 on 28-09-2025.

Login Screen:



The login form is titled "Login". It contains the following fields and buttons:

- Email: archana2@gmail.com
- Password: (masked with dots)
- Login button (blue)

Reminder popup:

127.0.0.1:5500 says

Overdue vaccines: Samruddhi - BCG, Samruddhi - OPV-0, Samruddhi - HepB-1, Samruddhi - DPT-1, Samruddhi - OPV-1, Samruddhi - HepB-2, Samruddhi - DPT-2, Samruddhi - OPV-2, Samruddhi - HepB-3

OK

Appointments

VacciCare+

HomeAdminHospitalParentAboutFeedback

Admin Portal

Dashboard

ChildrenAppointmentsHospitalsFeedback

Export CSV

ID	Child	Parent	Hospital	Date	Time	Status
4	prerna	Usha	shree lifecare hospital	2025-09-16	22:44	pending

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Admin Manual

Admin login:

VacciCare+

HomeAdminHospitalParentAboutFeedback

Admin Portal

Admin Login

Username

admin

Password

admin123

Login

Default: admin / admin123

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Admin Portal:

VacciCare+

HomeAdminHospitalParentAboutFeedback

Admin Portal

Dashboard

Logout

ChildrenAppointmentsHospitalsFeedback

Export CSV

ID	Name	Parent	DOB	Vaccines
2	prerna	Usha	2025-04-28	
6	Samruddhi	Archana Fartade	2025-07-11	
8	Chitra	Varsha Bhandalkar	2025-08-04	
10	Ansh	Karuna Khandekar	2025-11-04	
12	Vaishnavi	Sunita Kale	2025-04-04	
14	Tushar	Savita Dhonge	2025-05-05	
16	Avinash	Anamika Kale	2025-05-06	
18	Padmakar	Asha Salunke	2025-06-05	
20	Premraj	Karina Fartade	2025-09-08	

Hospital Module:

Hospital Registration & Login:

VacciCare+

HomeAdminHospitalParentAboutFeedback

Hospital Portal

Register Hospital

Hospital Name

Dignity9 Hospital

Email

dignity6@gmail.com

Password

Register

Admin must approve before you can manage appointments.

Hospital Login

Email

dignity6@gmail.com

Password

hospital123

Login

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VacciCare+

HomeAdminHospitalParentAboutFeedback

Admin Portal

Dashboard

Logout

ChildrenAppointmentsHospitalsFeedback

ID	Name	Email	Approved	Action
3	shree lifecare hospital	hosp1@gmail.com	Yes	Revoke
21	Global Multispeciality Hospital	global2@gmail.com	Yes	Revoke
22	Jeevan Rakshak Hospital	jeevan3@gmail.com	No	Approve
23	Care Multispeciality Hospital	care4@gmail.com	No	Approve
24	Dignity9 Hospital	dignity6@gmail.com	No	Approve
25	Fortune Multispeciality Hospital	fortune7@gmail.com	No	Approve
26	Pulse Speciality Hospital	pulse8@gmail.com	No	Approve
27	Metro Superspeciality Hospital	metro9@gmail.com	No	Approve

Feedback:

VacciCare+

HomeAdminHospitalParentAboutContact

Feedback

Feedback Form

Name

Archana fartade

Contact

1234567890

Rating

★

★

★

★

★

Comments

I like this website . This helps me to find good hospital and give vaccination reminders.

Submit

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VacciCare+

HomeAdminHospitalParentAboutFeedback

Admin Portal

Dashboard

Logout

Children

Appointments

Hospitals

Feedback

Name	Contact	Rating	Comments	Date
Archana fartade	1234567890	4	I like this website . This helps me to find good hospital and give vaccination reminders.	2025-09-28
Usha Fartade	1234567890	3	this website is very useful..	2025-09-28
Karina Khandekar	9876544329	4	help me to find clean neat hospital..	2025-09-28
Rishika Kale	987567890	3	I like this website . give vaccination reminders.	2025-09-28
Varsha Bhandalkar	984567890	3	good website	2025-09-28
Ananad Kale	934567890	5	I like this website .	2025-09-28
Swarali Avachat	1234567988	4	This helps me to find good hospital and give vaccination reminders.	2025-09-28

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Testing and Implementation:

1. Testing Strategies

To ensure the VacciCare+ web application is reliable, secure, and user-friendly, a systematic testing strategy is adopted. The strategy covers different testing levels and methods to verify both functional and non-functional requirements.

- . Unit Testing: Tests individual modules (like login or registration) to ensure each works correctly.
- Integration Testing: Checks interactions between modules to verify data flows correctly.
- System Testing: Tests the complete system against functional and non-functional requirements.
- User Acceptance Testing (UAT): End-users test the system to ensure it meets their needs.
- Regression Testing: Re-tests the system after changes to ensure existing functionality is not broken.

2. Testing Techniques Used

1. Black Box Testing:

- Tests the application's functionality without considering internal code.
- Example: Verifying user login, vaccine booking, and notification features.

2. White Box Testing:

- Tests internal code logic and paths of modules.
- Example: Checking functions for data validation, appointment scheduling, and database updates.

3. Manual Testing:

- Test cases are executed manually to verify system behavior.
- Example: Users booking appointments and checking records manually.

4. Automation Testing (Optional):

- Uses scripts or tools to test repetitive tasks efficiently.
- Example: Automated email/SMS notification checks.

3. Implementation Approach

Phase 1: Planning & Requirement Analysis:

- Study user requirements and define system specifications.
- Prepare the design and architecture of the application.

Phase 2: System Design & Development:

- Design the database, user interface, and system modules.
- Develop core modules like registration, login, vaccine management, appointment booking, and notifications.

Phase 3: Testing :

- Perform unit testing, integration testing, system testing, regression testing, and user acceptance testing (UAT).
- Identify and fix errors or bugs.

Phase 4: Deployment :

- Deploy the application on a live web server.
- Ensure data migration and proper server configuration.

Phase 5: Training & Documentation :

- Train admins and users to use the system efficiently.
- Provide user manuals and system documentation.

Phase 6: Maintenance & Support :

- Monitor system performance and security.
- Apply updates, enhancements, and bug fixes as required.

Drawbacks and Limitations

1. Internet Dependency:

- The system requires a stable internet connection. Users cannot access it offline.

2. Technical Knowledge Required:

- Users with limited computer or smartphone skills may face difficulties navigating the system.

3. Server Downtime:

- In case of server failure or maintenance, the system may be temporarily unavailable.

4. Limited Offline Functionality:

- Features like notifications, reminders, or record updates may not work offline.

Proposed Enhancements

1. Mobile Application:

- Develop a dedicated mobile app for Android and iOS to improve accessibility and user experience.

2. Offline Functionality:

- Enable limited offline features, such as viewing vaccination records without internet access.

3. Integration with Health Portals:

- Sync with government or hospital health portals for real-time vaccination data updates.

4. Automated Reminders via Multiple Channels:

- Include SMS, email.

Conclusion :

The VacciCare+ system provides an efficient and user-friendly solution for managing vaccination records and schedules. It simplifies the process for both users and administrators by offering online registration, appointment booking, reminders, and record maintenance.

The system enhances accessibility, reduces manual errors, and improves overall healthcare management. With proposed future enhancements like mobile apps, offline functionality, and advanced analytics, VacciCare+ has the potential to become a comprehensive platform for vaccination management, contributing significantly to public health awareness and safety.

Bibliography

- W3Schools – HTML, CSS & JavaScript Tutorials:
- https://www.w3schools.com/references/index.php?utm_source=chatgpt.com
- PHP Official Documentation: <https://www.php.net/docs.php>
- MySQL Documentation: <https://dev.mysql.com/doc/>
- Object Management Group (OMG) – UML Documentation:
- <https://www.omg.org/spec/UML/>
- Tutorialspoint – Web Development and PHP Tutorials:
- <https://www.tutorialspoint.com/>
- YouTube – Web Development & Full Stack Projects:
<https://youtube.com/>
- Wang, J. (2018). Implementing Geolocation Services for Improved Accuracy in Tracking Systems. Journal of Computer Science and Technology, 33(4), 729–738.
- Kumar, V., & Rani, S. (2020). Development of an Online Examination System. International Journal of Computer Applications, 175(6), 12–19.